GLOBAL DEVELOPMENT ANALYSIS

PROJECT REPORT

Data Cleaning • Feature Engineering • Visualisation & Insights

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1. Project Overview

This report documents the complete end-to-end data analysis pipeline conducted on the World Bank dataset. The project covers raw data ingestion, systematic data cleaning, feature engineering, database storage, and multi-dimensional visualisation with actionable insights. The goal is to understand socioeconomic development patterns across Asian sub-regions using a wide array of development indicators.

1.1 Objectives

- Clean and prepare raw World Bank data for analysis
- Engineer meaningful composite indices (health, education, gender, environment, development)
- Store processed data in a PostgreSQL database for query-based analysis
- Generate visualisations to uncover patterns, trends, and regional disparities
- Derive actionable insights on economic growth, health outcomes, education, and gender equality

1.2 Tools & Technologies

Tool / Library	Purpose
Python (Pandas, NumPy)	Data manipulation and cleaning
Matplotlib / Seaborn	Data visualisation,Streamlit
PostgreSQL + SQLAlchemy	Data storage and SQL querying
Jupyter Notebook	Interactive development environment

2. Data Cleaning

The raw dataset (raw_messy_data.csv) required extensive preprocessing before analysis. This section details every transformation applied to ensure data quality and analytical reliability.

```python
import numpy as np
import pandas as pd
```

✓ 1.1s Python

```python
df = pd.read_csv('../Data/raw_messy_data.csv')
```

✓ 0.5s Python

2.1 Initial Exploration

The dataset was first loaded and inspected using standard Pandas methods (head, shape, describe, info) to understand its structure, data types, and the extent of missing values. This gave an initial picture of which columns needed attention.

```python
df.head()
```

✓ 0.0s Python

	economy	Country	year	gdp_per_capita	gdp_growth	fdi_inflows	unemployment	poverty_headcount	gini_index	life_expectancy	...
0	AFG	Afghanistan	YR2000	174.930991	NaN	0.004828	7.897	NaN	NaN	55.005	...
1	AFG	Afghanistan	YR2001	138.706822	-9.431974	0.024169	7.973	NaN	NaN	55.511	...
2	AFG	Afghanistan	YR2002	178.954088	28.600001	1.306950	7.867	NaN	NaN	56.225	...
3	AFG	Afghanistan	YR2003	198.871116	8.832278	1.278493	7.844	NaN	NaN	57.171	...
4	AFG	Afghanistan	YR2004	221.763654	1.414118	3.577104	7.794	NaN	NaN	57.810	...

5 rows × 28 columns

```python
df.shape
```

✓ 0.0s Python

```
(1128, 28)
```

```python
df.describe()
```

Python

	gdp_per_capita	gdp_growth	fdi_inflows	unemployment	poverty_headcount	gini_index	life_expectancy	infant_mortality	hospital_beds
count	1098.000000	1097.000000	1079.000000	1127.000000	363.000000	363.000000	1128.000000	1128.000000	966.000000
mean	10879.168971	4.771827	3.758433	6.272689	11.375758	35.203857	72.173427	23.999734	3.055476
std	16444.309719	6.103189	5.668861	4.876761	16.353972	4.904732	5.859500	19.829006	2.702960
min	136.925716	-36.656780	-37.172653	0.100000	0.000000	25.100000	55.005000	1.700000	0.200000
25%	1220.249354	2.490949	0.737073	3.067500	0.700000	31.500000	67.921500	8.700000	1.340000
50%	3578.411232	5.045124	2.185494	4.693000	3.400000	35.300000	72.430000	18.000000	2.170000
75%	11382.938567	7.328374	4.708120	8.553500	16.500000	38.800000	76.278250	33.625000	4.047500
max	108470.387023	58.078101	55.072877	27.465000	93.800000	47.700000	84.560000	110.100000	14.670000

rows × 24 columns

```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 1128 entries, 0 to 1127
Data columns (total 28 columns):
 #   Column                      Non-Null Count  Dtype
---  ------                      --------------  -----
 0   economy                     1128 non-null   str
 1   Country                     1128 non-null   str
 2   year                        1128 non-null   str
 3   gdp_per_capita              1098 non-null   float64
 4   gdp_growth                  1097 non-null   float64
 5   fdi_inflows                 1079 non-null   float64
 6   unemployment                1127 non-null   float64
 7   poverty_headcount           363 non-null    float64
 8   gini_index                  363 non-null    float64
 9   life_expectancy             1128 non-null   float64
 10  infant_mortality            1128 non-null   float64
 11  hospital_beds               966 non-null    float64
 12  population                  1128 non-null   float64
 13  literacy_rate               324 non-null    float64
 14  school_enrolment_primary    930 non-null    float64
 15  school_enrolment_secondary  820 non-null    float64
 16  education_expenditure       727 non-null    float64
 17  electricity_access          1119 non-null   float64
 18  renewable_energy            1038 non-null   float64
 19  forest_area                 1128 non-null   float64
...
 26  internet_users              1077 non-null   float64
 27  sub_region                  1128 non-null   str
dtypes: float64(24), str(4)
memory usage: 246.9 KB
```

2.2 Year Column Formatting

The year column contained string values prefixed with 'YR' (e.g., 'YR2000'). The prefix was stripped using lstrip('YR') and the values were cast to integer, enabling chronological sorting and time-series analysis.

```python
df['year']=df["year"].str.lstrip('YR').astype(int)
```

2.3 Handling Missing & Invalid Values

Each column was treated individually based on its nature and the extent of missing data. The table below summarises the strategy applied per column group.

Column(s)	Strategy	Rationale
gdp_per_capita	Impute with country-wise median; drop Korea Dem.	Country median preserves economic context; NK had insufficient data
gdp_growth	Drop rows with null values	Too few data points to meaningfully impute growth rates
fdi_inflows	Impute with country-wise median	Country-level median reflects typical investment patterns
unemployment	Drop rows with null values	Sparse data; imputation would introduce noise

poverty_headcount / gini_index	Drop entire columns	Excessive missing values (>50%) make them unreliable
hospital_beds	Impute with sub-region median	Sub-region reflects similar healthcare infrastructure
literacy_rate / female_literacy	Drop entire columns	Too sparse to be analytically useful
school_enrolment (primary & secondary)	Impute with sub-region median	Sub-regional patterns are most representative
education_expenditure	Impute with sub-region median	Consistent with regional policy spending patterns
renewable_energy, clean_water_access	Impute with sub-region median	Environment indicators vary by geography/sub-region
women_parliament, gender_parity	Impute with sub-region median	Gender metrics are regionally correlated
internet_users	Impute with sub-region median	Digital access closely follows regional infrastructure

```python
df.drop(df[df['Country'] =="Korea, Dem. People's Rep."].index,inplace=True)
```

Python

```python
df['gdp_per_capita']=df.groupby('Country')['gdp_per_capita'].transform(lambda x : x.fillna(x.median()))
```

Python

```python
df['gdp_per_capita'].isnull().sum()
```

Python

```
np.int64(0)
```

```python
df['gdp_per_capita']=df['gdp_per_capita'].round(2)
```

Python

```python
df.head()
```

Python

	economy	Country	year	gdp_per_capita	gdp_growth	fdi_inflows	unemployment	poverty_headcount	gini_index	life_expectancy	...	r
0	AFG	Afghanistan	2000	174.93	NaN	0.004828	7.897	NaN	NaN	55.005	...	
1	AFG	Afghanistan	2001	138.71	-9.431974	0.024169	7.973	NaN	NaN	55.511	...	
2	AFG	Afghanistan	2002	178.95	28.600001	1.306950	7.867	NaN	NaN	56.225	...	
3	AFG	Afghanistan	2003	198.87	8.832278	1.278493	7.844	NaN	NaN	57.171	...	
4	AFG	Afghanistan	2004	221.76	1.414118	3.577104	7.794	NaN	NaN	57.810	...	

5 rows × 28 columns

2.4 Rounding & Final Validation

All numeric columns were rounded to 3 decimal places for consistency. A final isnull().sum() check confirmed zero remaining nulls in the cleaned dataset. The country column was renamed to lowercase (country) for database convention alignment.

> *Key Insight: After cleaning, the dataset retained all key analytical columns with zero null values, ensuring a robust foundation for downstream analysis.*

```
df.isnull().sum()
```

```
economy                     0
Country                     0
year                        0
gdp_per_capita              0
gdp_growth                  0
fdi_inflows                 0
unemployment                0
life_expectancy             0
infant_mortality            0
hospital_beds               0
population                  0
school_enrolment_primary    0
school_enrolment_secondary  0
education_expenditure       0
electricity_access          0
renewable_energy            0
forest_area                 0
clean_water_access          0
female_labour               0
women_parliament            0
maternal_mortality          0
gender_parity               0
internet_users              0
sub_region                  0
dtype: int64
```

3. Feature Engineering

Six new composite features and four categorical classification columns were engineered to enable deeper multi-dimensional analysis.

3.1 Derived Numeric Features

Feature	Formula	What It Measures
gdp_total	*gdp_per_capita x population*	Total economic output of a country
fdi_per_capita	*(fdi_inflows x 1B) / population*	Foreign investment normalized per person
health_index	*(life_expectancy / 100) + (1 / infant_mortality)*	Combined health outcome score
education_index	*(primary_enrolment + secondary_enrolment) / 2*	Average school participation rate

environment_index	*(renewable_energy + forest_area + clean_water) / 3*	Environmental sustainability score
gender_index	*(female_labour + women_parliament + gender_parity) / 3*	Gender equality composite score
development_index	*(electricity + clean_water + internet + education_index) / 4*	Overall access & infrastructure score

3.2 Categorical Classifications

Four binned categorical columns were created using pd.cut() to enable grouped analysis and cross-tabulation.

Category Column	Labels
development_category	Very Poor | Underdeveloped | Developing | Moderate | Developed | Highly Developed
life_expectancy_category	Very Low | Low | Moderate | High | Very High
environment_category	Critical | Poor | Moderate | Good | Excellent
gdp_category	Low Income | Lower Middle | Upper Middle | High Income | Very High Income
gender_category	Very Poor | Poor | Moderate | Good | Excellent

```python
df['health_index']=(df['life_expectancy']/100) + (1/df['infant_mortality'])
```

Python

```python
df['education_index']=(df['school_enrolment_primary'] + df['school_enrolment_secondary'])/2
```

Python

```python
df['environment_index']=(df['renewable_energy'] + df['forest_area'] + df['clean_water_access'])/3
```

Python

```python
df['gender_index']=(df['female_labour'] + df['women_parliament'] + df['gender_parity']) / 3
```

Python

```python
df['development_index']=(df['electricity_access'] + df['clean_water_access']+df['internet_users']+df['education_index'])/4
```

Python

```python
df['development_category']=pd.cut(df['development_index'],
        bins=[0,30,50,70,90,100,110],
        labels=['Very Poor','Underdeveloped','Developing','Moderate','Developed','Highly Developed']
    )
```

Python

The cleaned and engineered dataset was exported to world_bank_cleaned.csv and loaded into a PostgreSQL database (table: World_Bank) for subsequent SQL-based analysis.

4. Visualisation & Insights

All visualisations were generated by querying the PostgreSQL database using SQLAlchemy and rendering results with Seaborn and Matplotlib. Analyses are grouped thematically below.

4.1 Economic Analysis

4.1.1 Development Category Distribution

A pie chart was used to show the proportion of countries in each development category. The majority of countries fall in moderate to developing stages, confirming significant global inequality. Very few nations qualify as 'Highly Developed'.

> *Key Insight: Global development remains uneven — most countries are still in the moderate-to-developing band, highlighting the scale of work needed to close the development gap.*

4.1.2 Countries by GDP Category

A bar chart visualised the count of countries per GDP income bracket. The distribution was heavily skewed towards lower-income categories, with only a handful of nations achieving very high income status.

> *Key Insight: Global wealth is highly concentrated: the vast majority of countries remain in lower income tiers, underscoring persistent economic inequality across nations.*

4.1.3 Average GDP Per Capita by Sub-Region

A comparative bar chart revealed that East Asia and West Asia lead significantly in average GDP per capita, while South Asia and Central Asia lag behind. Differences in industrialisation, oil wealth (West Asia), and technological advancement explain much of this gap.

4.1.4 Countries with Most Frequent Negative GDP Growth

A bar chart of the top 10 countries with the highest count of negative GDP growth years identified conflict-affected nations (Syrian Arab Republic, Yemen) and resource-dependent economies (Brunei, Kuwait) as the most vulnerable. This highlights the importance of economic diversification and political stability.

4.1.5 Global Average GDP Growth Over Time

A line chart with a zero-growth reference line showed that global GDP growth has generally remained positive, with a notable peak in the mid-2000s, a sharp contraction in 2009 (global financial crisis), and the deepest drop in 2020 due to the COVID-19 pandemic. Recovery post-2020 has been gradual and uneven.

4.1.6 FDI Inflows by Sub-Region

Southeast Asia attracts the highest average FDI inflows, cementing its role as a global manufacturing and investment hub. South Asia shows the lowest inflows, suggesting untapped potential and barriers such as regulatory complexity and infrastructure gaps.

4.1.7 FDI vs. GDP Growth Relationship

A regression scatter plot with correlation analysis showed a weak-to-moderate positive relationship between FDI inflows and GDP growth. The significant variability and outliers confirm

that FDI alone does not guarantee economic growth — political stability, infrastructure, and domestic policy all play critical roles.

4.1.8 GDP Per Capita Distribution by Income Category

A box plot with strip overlay showed that GDP per capita increases substantially from low-income to very high-income groups, with increasing variance at higher income levels. The Very High Income category contains notable outliers representing exceptionally wealthy economies.

4.2 Health Analysis

4.2.1 Life Expectancy by Sub-Region

A violin plot showed that East Asia and West Asia have the highest life expectancy distributions, while South Asia and Central Asia trail behind. Healthcare quality, income, nutrition, sanitation, and public health investment all contribute to these differences.

4.2.2 Infant Mortality by Sub-Region

South Asia recorded the highest average infant mortality rate, while East Asia had the lowest. This regional disparity reflects inequalities in maternal care, nutrition, healthcare infrastructure, and sanitation.

4.2.3 GDP Per Capita vs. Life Expectancy

A regression plot confirmed a clear positive relationship: as income rises, life expectancy increases — especially at lower income levels. At higher GDP levels, the relationship shows diminishing returns, where healthcare quality and lifestyle factors play a larger role.

> *Key Insight: Economic growth strongly improves health outcomes, but above a certain income threshold, other factors like healthcare system quality and lifestyle become the dominant drivers of longevity.*

4.2.4 Life Expectancy Trends Over Time (Selected Countries)

A multi-line chart for India, China, Japan, Nepal, Israel, Pakistan, and Saudi Arabia showed universal improvement over time. Japan and Israel consistently maintain the highest life expectancy, while India, Pakistan, and Nepal show steady progress from lower baselines.

4.3 Education Analysis

4.3.1 Secondary School Enrolment by Sub-Region

East Asia has the highest average secondary enrolment, while South Asia has the lowest. Poverty, limited infrastructure, and social factors create barriers to continuing education past primary level in lower-performing sub-regions.

4.3.2 Education Index vs. GDP Per Capita

A regression plot confirmed a positive correlation: countries with higher education levels tend to have stronger economies. Variability in the data confirms that while education is a key driver, governance, natural resources, and industrialisation also influence economic performance.

4.3.3 Primary vs. Secondary Enrolment Gap

A grouped bar chart comparing primary and secondary enrolment by sub-region showed that primary enrolment is consistently high across all regions, but secondary enrolment drops significantly — most sharply in South Asia. This dropout pattern highlights economic and social constraints that prevent students from continuing their education.

4.3.4 Education Expenditure Distribution

A box plot with strip overlay showed that Central and West Asia maintain higher and more consistent education expenditure, while South Asia has lower spending. Southeast Asia shows high variability, indicating inconsistent investment across countries within the region.

4.4 Environment Analysis

4.4.1 Top 10 Countries by Renewable Energy Usage

Bhutan and Nepal topped the chart, relying heavily on hydroelectric power. Larger and more industrialised economies tend to depend more on fossil fuels, highlighting the inverse relationship between industrialisation scale and renewable energy share.

4.4.2 GDP vs. Renewable Energy Relationship

A regression scatter plot revealed a negative relationship: wealthier countries tend to use less renewable energy as a share of total energy. This suggests that higher-income nations rely more heavily on fossil fuels, highlighting a gap in sustainable economic development.

> *Key Insight: Richer countries are often bigger polluters. Economic prosperity has not yet translated into greater renewable energy adoption for most high-income nations.*

4.4.3 Clean Water Access Over Time

A line chart showed a steady and consistent improvement in global clean water access over the observed period, reflecting positive policy outcomes and infrastructure investment worldwide.

4.4.4 GDP vs. Renewable Energy Quadrant Analysis

A scatter plot with median reference lines divided countries into four quadrants: Rich & Green, Rich & Polluting, Poor & Green, and Poor & Polluting. Most countries fall into the 'Poor but Green' or 'Rich but Less Green' categories. Very few nations combine high income with high renewable energy dependence, illustrating the sustainability challenge.

4.5 Gender Equality Analysis

4.5.1 Female Labour Participation by Sub-Region

A violin plot showed that Southeast and East Asia have notably higher female labour participation, while South and West Asia exhibit lower rates. Cultural norms, policy barriers, and differing employment opportunities explain the regional divergence.

4.5.2 Gender Index vs. GDP Per Capita

A regression plot identified a slight positive relationship between gender equality and per capita income. Countries with better gender equity tend to be wealthier, though the relationship is weak, indicating that economic growth alone does not guarantee gender parity.

4.5.3 Women's Parliamentary Representation Over Time

A line chart showed a gradual but consistent upward trend in women's parliamentary representation globally, reflecting slow but measurable progress toward political gender equality.

4.6 Composite & Overall Development Analysis

4.6.1 Correlation Heatmap of Key Indicators

A correlation heatmap confirmed strong positive relationships between GDP per capita, life expectancy, and development index. Infant mortality showed strong negative correlations with these indicators, confirming that economic and health development are deeply interconnected.

4.6.2 Development Balance Across Sub-Regions

A grouped bar chart comparing health, education, gender, and environment indices across sub-regions showed that East Asia is the most balanced and highest-performing region. South Asia consistently scores lowest across all dimensions. West Asia shows a notable gap in gender equality despite decent education scores.

4.6.3 Development Index Heatmap

A heatmap with annotated scores for all sub-regions and index types provided a visual summary of regional development. East Asia leads comprehensively, while South Asia shows the most room for improvement across all development pillars.

> *Key Insight: No single region excels uniformly across all development dimensions. East Asia comes closest to balanced development, while South Asia's low scores across health, gender, and development index signal the need for urgent, multi-sectoral policy interventions.*

4.7 Interactive Dashboard (Streamlit)‘

An interactive dashboard was developed using Streamlit to transform the static analysis into a dynamic, user-driven exploration tool. The dashboard enables users to select countries, adjust time ranges, and analyze development indicators in real time.
The interface is designed with a clear separation between:

- **Focus Country**: Used for KPI indicators and ADIS score calculation
- **Compare Countries**: Used for multi-country visualizations and trend analysis

This ensures accurate metric computation while allowing flexible comparison across countries.
Key Features:

- Dynamic filtering by country and year range
- KPI cards displaying key indicators (GDP, health, education, digital access)
- Custom **ADIS Score** representing overall development performance
- Multi-dimensional visualizations including line charts, bar charts, and scatter plots
- Bubble charts incorporating population as a third analytical dimension
- Real-time updates based on user input

Insight Delivery:
The dashboard enhances analytical capabilities by:

- Allowing users to identify trends over time
- Comparing development indicators across countries
- Understanding relationships between economic, social, and environmental variables

- Highlighting strengths and weaknesses through composite scoring

5. Key Conclusions

The following cross-cutting conclusions emerged from the complete analysis:

- Economic development strongly correlates with health and educational outcomes. Countries with higher GDP per capita consistently show better life expectancy, lower infant mortality, and higher school enrolment.
- Foreign Direct Investment has a weak-to-moderate positive effect on GDP growth. It is not a standalone growth engine — governance, infrastructure, and domestic policies are equally important.
- South Asia presents the most significant development challenges across all pillars: economy, health, education, gender, and environment.
- East Asia demonstrates the most balanced and advanced development profile, setting a regional benchmark.
- The COVID-19 pandemic caused the most severe disruption to global GDP growth in the dataset, with recovery remaining slow and uneven.
- Wealthy nations tend to rely more on fossil fuels, creating a tension between economic prosperity and environmental sustainability.
- Women's parliamentary representation and female labour participation are rising globally but remain uneven, particularly in South and West Asia.
- Access to clean water has improved consistently worldwide, representing one of the most positive global development trends in the data.